



Manajemen Proyek Sistem Informasi

PROJECT SCHEDULE MANAGEMENT

S1 SISTEM INFORMASI

FAKULTAS REKAYASA INDUSTRI

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Learning Objectives

- Understand the **importance of project schedules** and good project time management
- Define **activities** as the basis for developing project schedules
- Describe **how project managers use network diagrams** and dependencies to assist in activity sequencing
- Understand the **relationship between estimating resources and project schedules**
- Explain how **various tools and techniques help project managers perform activity duration estimating**
- Use a **Gantt chart for planning and tracking schedule information**, find the critical path for a project, and describe how critical chain scheduling and the Program Evaluation and Review Technique (PERT) affect schedule development
- Discuss **how reality checks and people issues** are involved in controlling and managing changes to the project schedule
- Describe **how project management software** can assist in project time management and review words of caution before using this software



Importance of Project Schedules

- Managers often cite **delivering projects on time as one of their biggest challenges**
- Time has **the least amount of flexibility**; it passes no matter what happens on a project
- **Schedule issues are the main reason for conflicts on projects**, especially during the second half of projects



Individual Work Styles and Cultural Differences Cause Schedule Conflicts

- One dimension of the **Meyers-Briggs Type Indicator** focuses on **peoples' attitudes toward structure and deadline**
- Some people prefer to **follow schedules** and meet deadlines **while others do not** (**J vs. P**)
- **Difference cultures** and even entire countries **have different attitudes** about schedules



Media Snapshot

- In contrast to the 2002 Salt Lake City Winter Olympic Games (see Chapter 4's Media Snapshot), planning and scheduling was very different for the 2004 Summer Olympic Games held in Athens, Greece
- Many articles were written before the opening ceremonies predicting that the facilities would not be ready in time; many people were pleasantly surprised by the amazing opening ceremonies, beautiful new buildings, and state-of-the-art security and transportation systems in Athens
- The Greeks even made fun of critics by having construction workers pretend to still be working as the ceremonies began



Project Schedule Management Processes

1. **Plan Schedule Management** : establishing the policies, procedures, and documentation for planning, developing, managing, executing and controlling the project Schedule
2. **Defining activities**: identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables
3. **Sequencing activities**: identifying and documenting the relationships between project activities
4. **Estimating activity resources**: estimating how many resources a project team should use to perform project activities
5. **Estimating activity durations**: estimating the number of work periods that are needed to complete individual activities
6. **Developing the schedule**: analyzing activity sequences, activity resource estimates, and activity duration estimates to create the project schedule
7. **Controlling the schedule**: controlling and managing changes to the project schedule

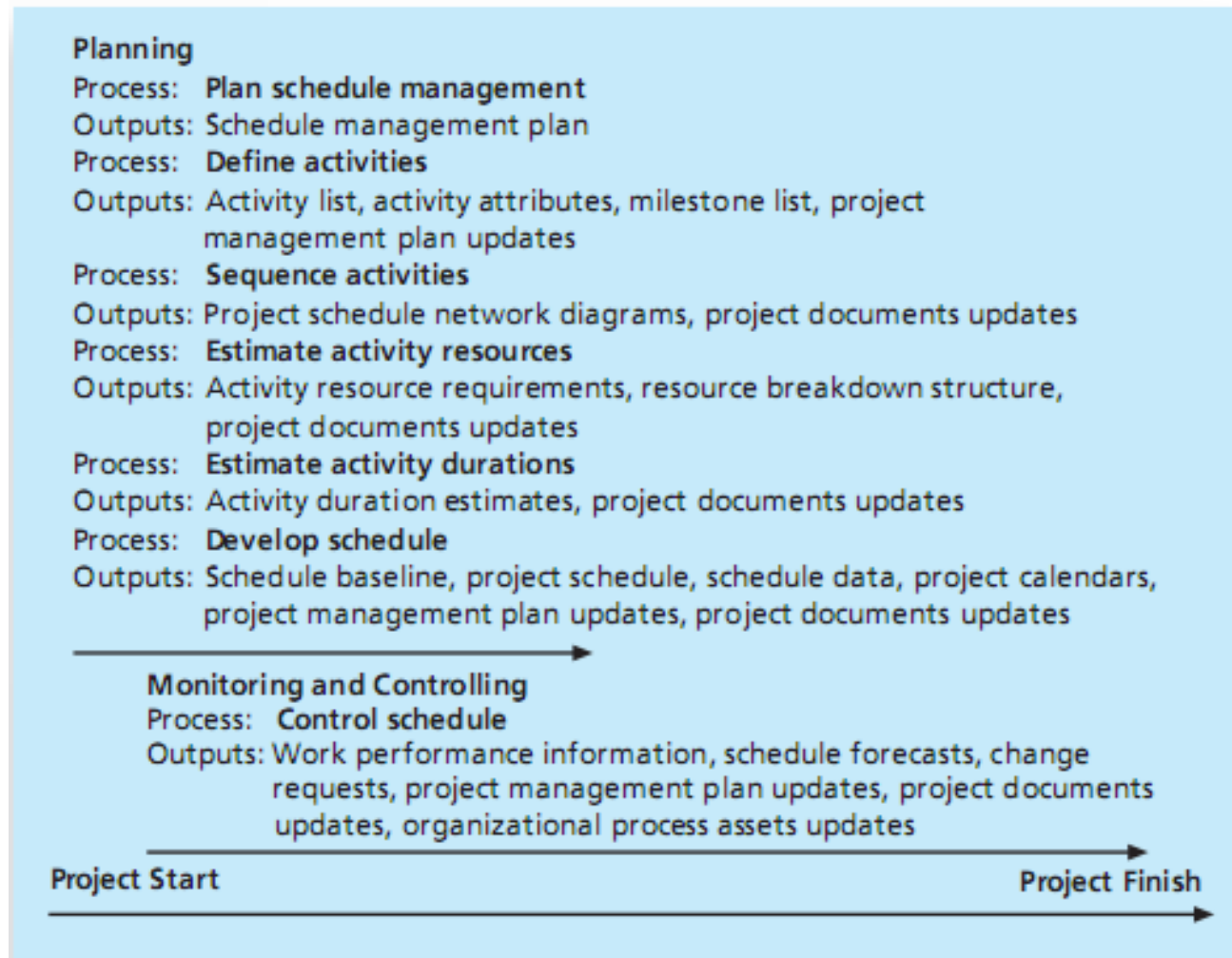




Project Management Process Groups and Knowledge Area Mapping

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
6. Project Time Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Resources 6.5 Estimate Activity Durations 6.6 Develop Schedule		6.7 Control Schedule	

Project Schedule Management Summary





1. PLANNING SCHEDULE MANAGEMENT

1. The schedule management plan, **like the scope management plan**, can be **informal and broad** or **formal and detailed**.
2. Schedule management plan includes the following information :
 - A. Project Schedule model development
 - B. Scheduling methodology and scheduling tools
 - C. Level accuracy and units of measure
 - D. Control thresholds
 - E. Rules of performance measurement
 - F. Reporting format
 - G. Process description





Plan Schedule Management ITTO





2. Defining Activities

- An **activity or task** is an element of work normally found on the work breakdown structure (WBS) that has an expected duration, a cost, and resource requirements
- **Activity definition** involves developing a more detailed WBS and supporting explanations to understand all the work to be done so you can develop realistic cost and duration estimates



Activity Lists and Attributes

- An **activity list** is a tabulation of activities to be included on a project schedule that includes:
 - The **activity name**
 - An **activity identifier** or number
 - A **brief description** of the activity
- **Activity attributes** provide more information such as predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions related to the activity

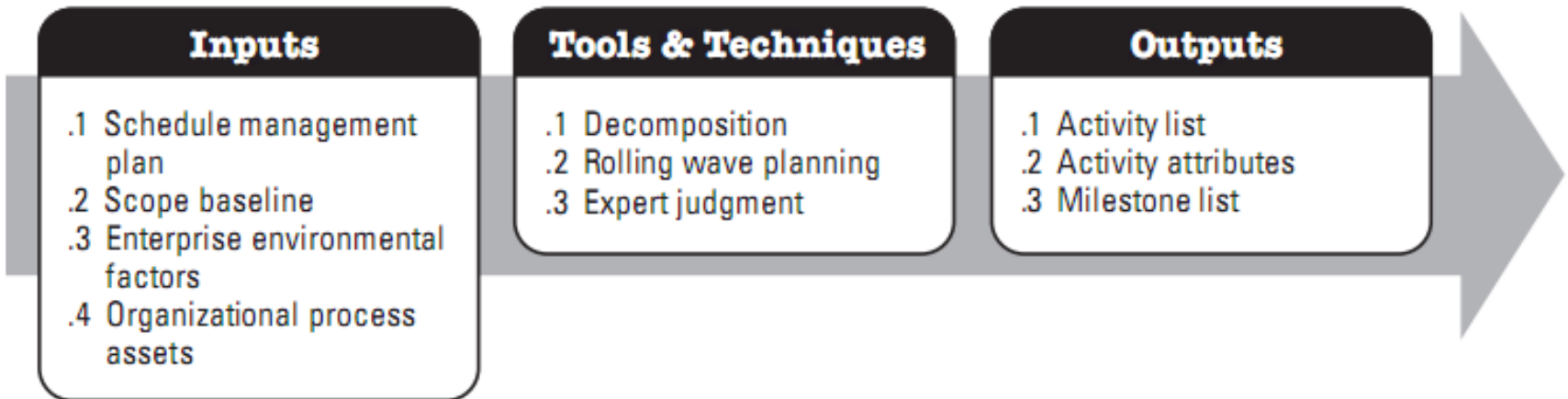


Milestones

- A **milestone** is a significant event that normally has no duration
- It **often takes several activities** and a lot of work **to complete a milestone**
- They're **useful tools for setting schedule goals** and monitoring progress
- Examples include obtaining customer sign-off on key documents or completion of specific products



Define Activities ITTO





3. Sequencing Activities

- Involves reviewing activities and determining dependencies
- A dependency or relationship is the sequencing of project activities or tasks
- You must determine dependencies in order to use critical path analysis



Three types of Dependencies

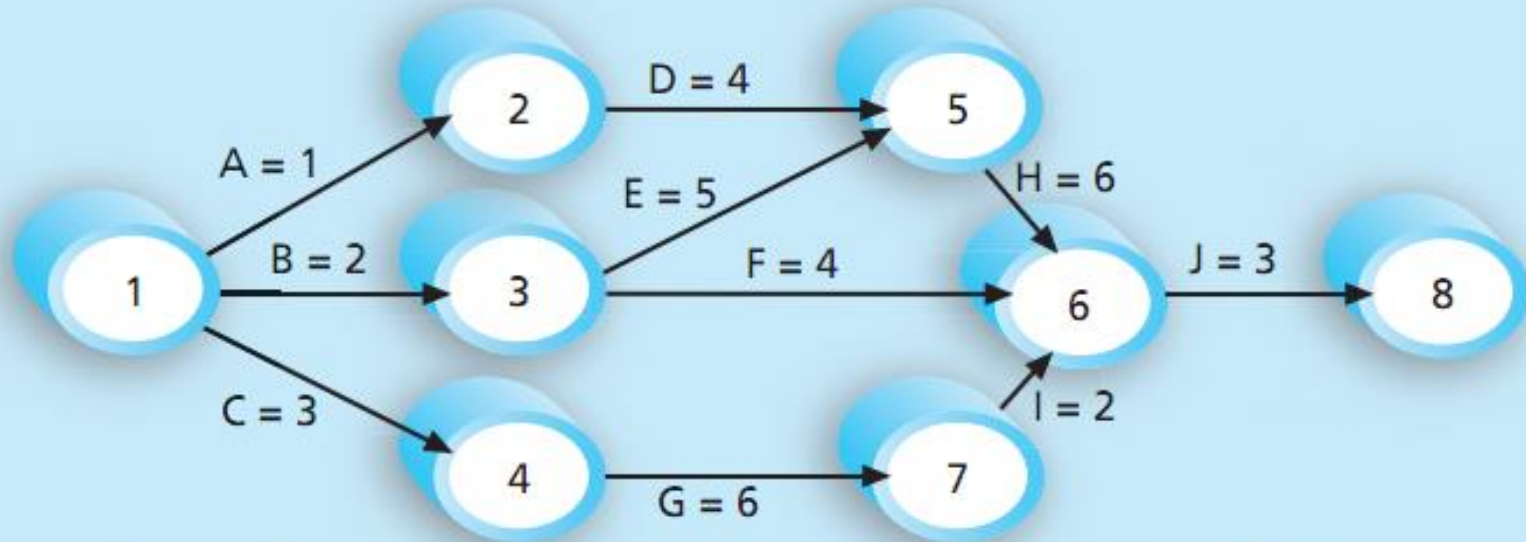
1. **Mandatory dependencies:** inherent in the nature of the work being performed on a project, sometimes referred to as **hard logic**
2. **Discretionary dependencies:** defined by the project team; sometimes referred to as **soft logic** and should be used with care since they may limit later scheduling options
3. **External dependencies:** involve **relationships between project and non-project activities**



Network Diagrams

- **Network diagrams** are the preferred technique for showing activity sequencing
- A **network diagram** is a schematic display of the logical relationships among, or sequencing of, project activities
- **Two main formats** are the arrow and precedence diagramming methods

Sample Activity-on-Arrow (AOA) Network Diagram for Project X



Note: Assume all durations are in days; A=1 means Activity A has a duration of 1 day.



Arrow Diagramming Method (ADM)

- Also called **activity-on-arrow (AOA)** network diagrams
- **Activities** are **represented by arrows**
- **Nodes or circles** are **the starting and ending points of activities**
- Can **only show finish-to-start** dependencies



Process for Creating AOA Diagrams

1. Find all of the activities that start at node 1. Draw their finish nodes and draw arrows between node 1 and those finish nodes. Put the activity letter or name and duration estimate on the associated arrow.
2. Continue drawing the network diagram, working from left to right. Look for bursts and merges. Bursts occur when a single node is followed by two or more activities. A merge occurs when two or more nodes precede a single node.
3. Continue drawing the project network diagram until all activities are included on the diagram that have dependencies.
4. As a rule of thumb, all arrowheads should face toward the right, and no arrows should cross on an AOA network diagram.





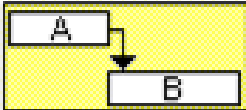
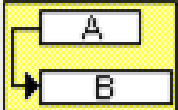
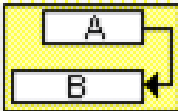
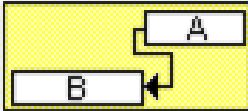
Precedence Diagramming Method (PDM)

- **Activities** are **represented by boxes**
- **Arrows** show **relationships between activities**
- **More popular than ADM method** and used by project management software
- **Better at showing different types of dependencies**

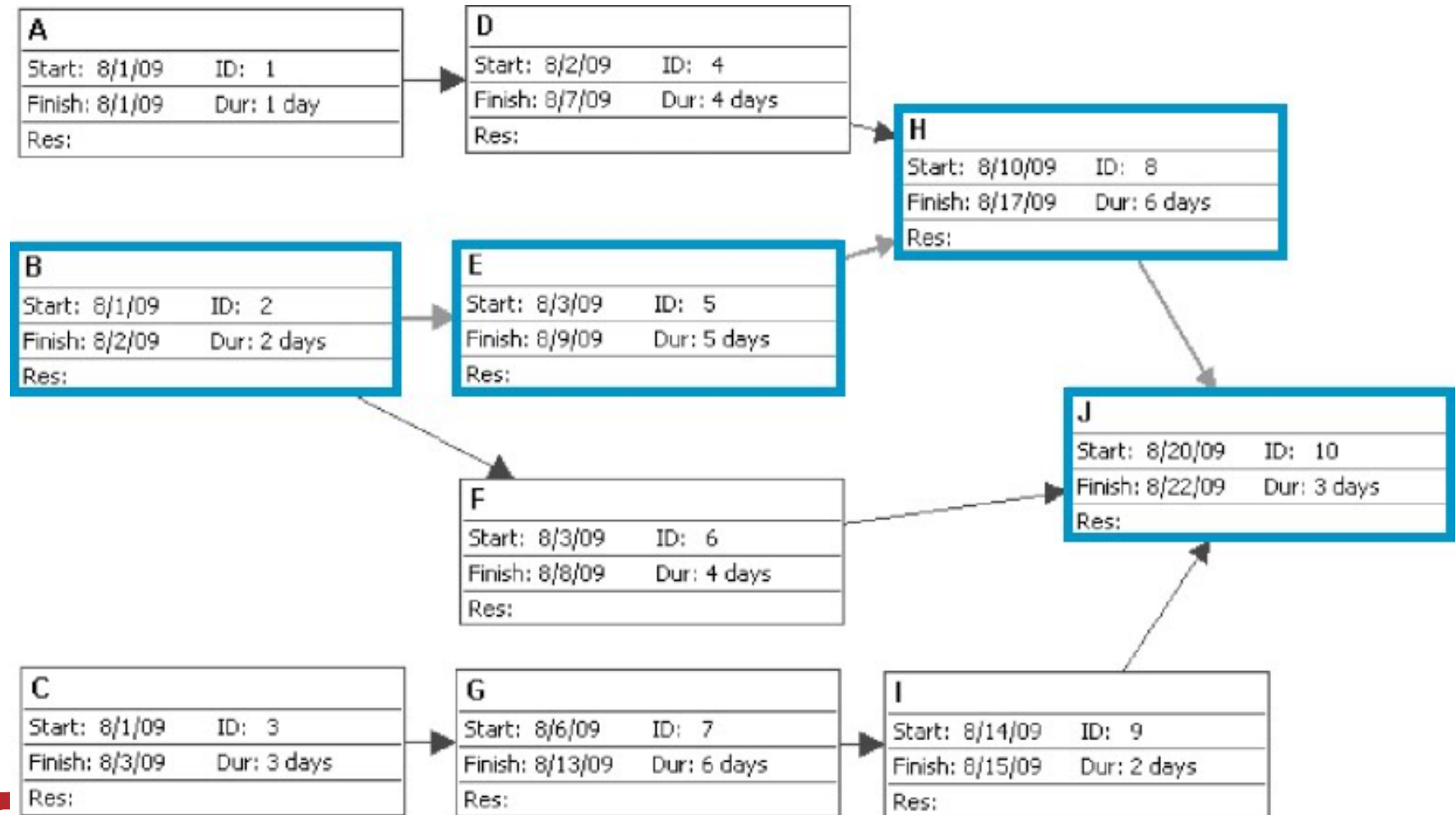
Task Dependency Types

Task dependencies

The nature of the dependencies between linked tasks. You link tasks by defining a dependency between their finish and start dates. For example, the "Contact caterers" task must finish before the start of the "Determine menus" task. There are four kinds of task dependencies in Microsoft Project:

Task dependency	Example	Description
Finish-to-start (FS)		Task (B) cannot start until task (A) finishes.
Start-to-start (SS)		Task (B) cannot start until task (A) starts.
Finish-to-finish (FF)		Task (B) cannot finish until task (A) finishes.
Start-to-finish (SF)		Task (B) cannot finish until task (A) starts.

Sample PDM Network Diagram



Sequence Activities ITTO

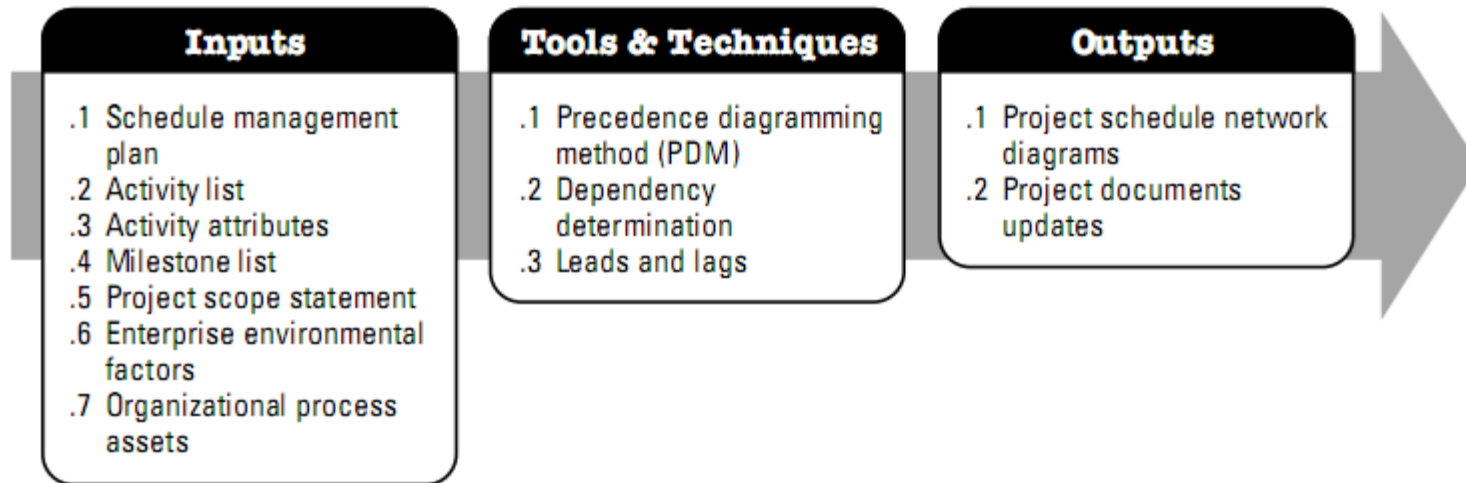


Figure 6-7. Sequence Activities: Inputs, Tools & Techniques, and Outputs



4. Estimating Activity Resources

- Before estimating activity durations, you must have a good idea of the **quantity and type of resources that will be assigned to each activity**
 - **resources** are **people**, **equipment**, and **materials**
- Consider **important issues** in estimating resources
 - **How difficult will it be** to do specific activities on this project?
 - **What is the organization's history** in doing similar activities?
 - Are the **required resources available**?
- A **resource breakdown structure** is a **hierarchical structure** that identifies the project's resources by category and type

Estimate Activity Resource ITTO

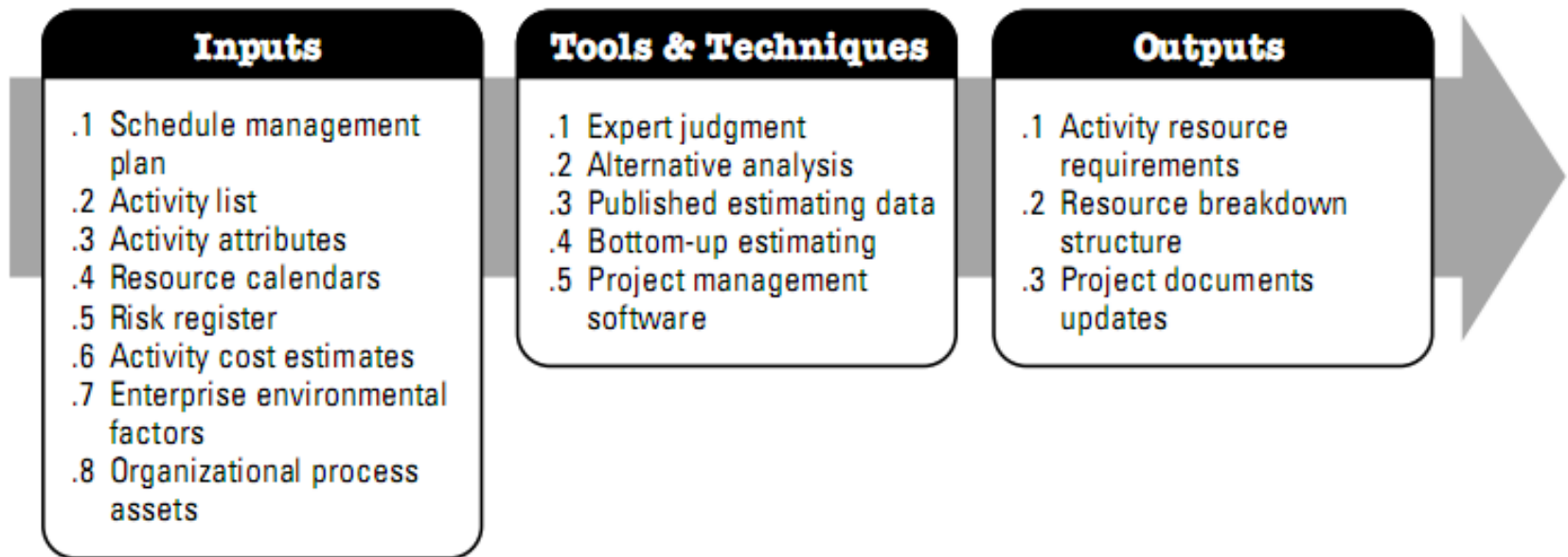


Figure 6-12. Estimate Activity Resources: Inputs, Tools & Techniques, and Outputs



5. Activity Duration Estimating

- **Duration** includes the **actual amount of time worked** on an activity plus elapsed time
- **Effort** is the **number of workdays or work hours** required to complete a task
- **Effort does not** normally **equal duration**
- People doing the work should **help create estimates**, and an **expert** should review them



Three-Point Estimates

- Instead of providing activity estimates as a discrete number, such as four weeks, it's often helpful to create a **three-point estimate**
 - An estimate that includes an **optimistic**, **most likely**, and **pessimistic estimate**, such as **three weeks for the optimistic**, **four weeks for the most likely**, and **five weeks for the pessimistic estimate**
- **Three-point estimates are needed** for **PERT** and **Monte Carlo simulations**

Estimate Activities Duration ITTO

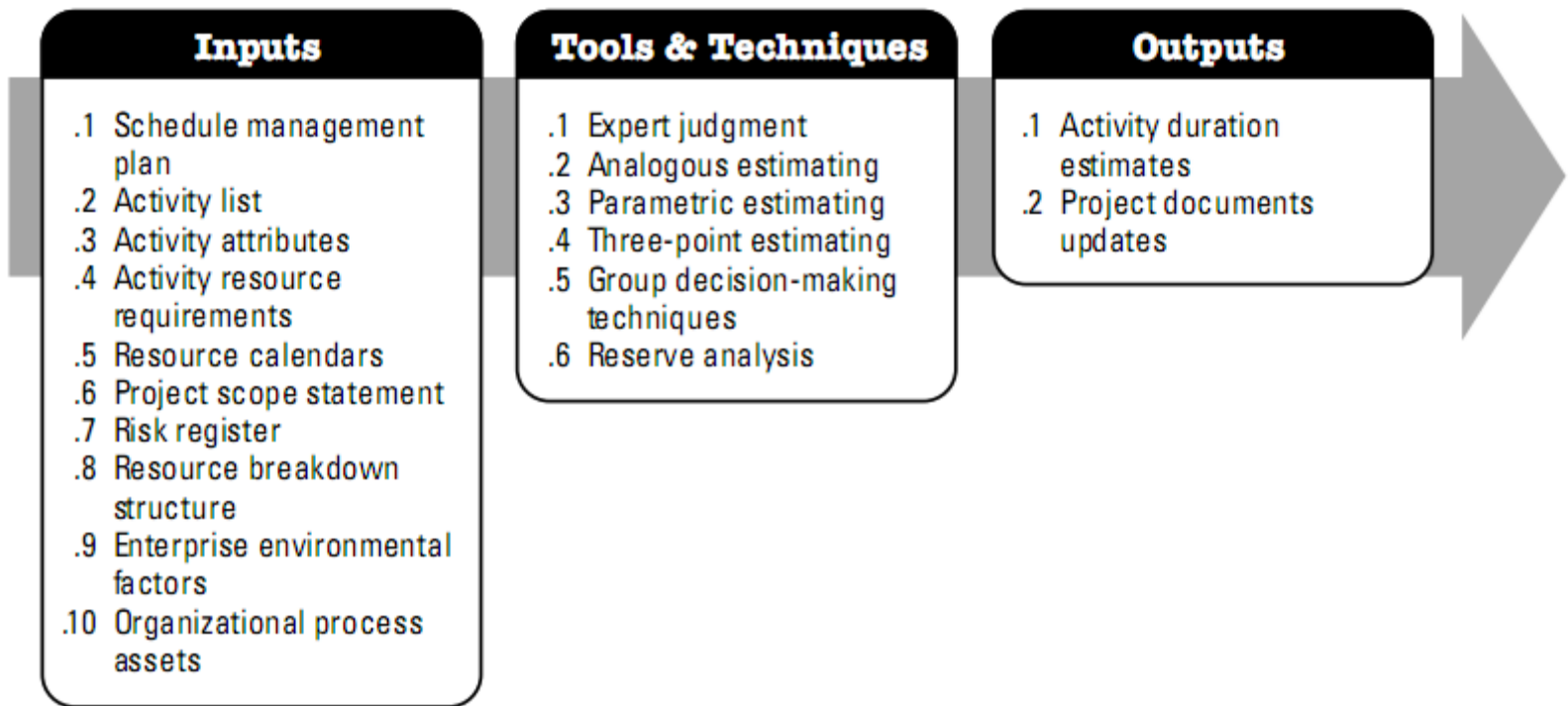


Figure 6-14. Estimate Activity Durations: Inputs, Tools & Techniques, and Outputs



6. Developing the Schedule

- Uses results of the other time management processes to **determine the start and end date** of the project
- **Ultimate goal** is to **create a realistic project schedule** that provides a basis for monitoring project progress for the time dimension of the project
- **Important tools** and techniques include **Gantt charts, critical path analysis, and critical chain scheduling, and PERT analysis**

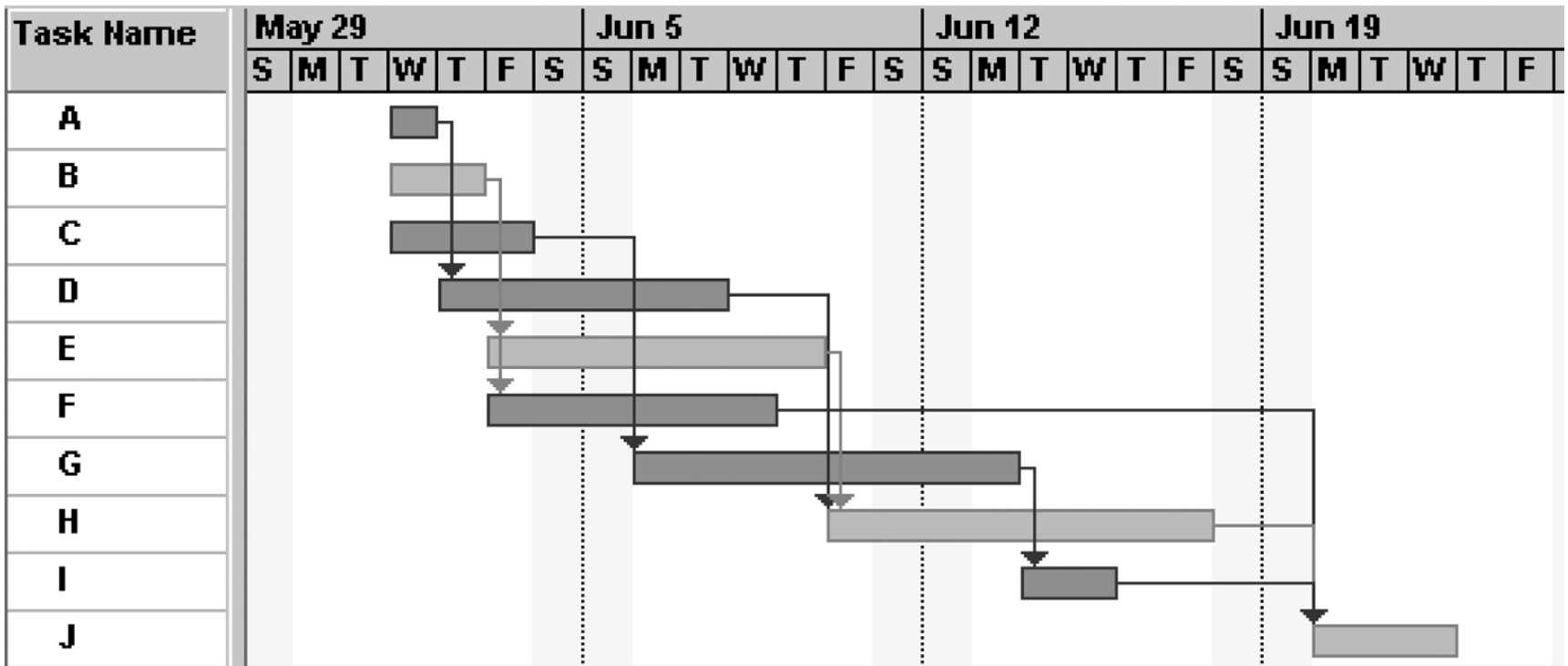


Gantt Charts

- **Gantt charts** provide a **standard format for displaying project schedule information** by listing project activities and their corresponding start and finish dates in a calendar format
- Symbols include:
 - **Black diamonds**: **milestones**
 - **Thick black bars**: **summary tasks**
 - **Lighter horizontal bars**: **durations of tasks**
 - **Arrows**: **dependencies between tasks**



Gantt Chart for Project X



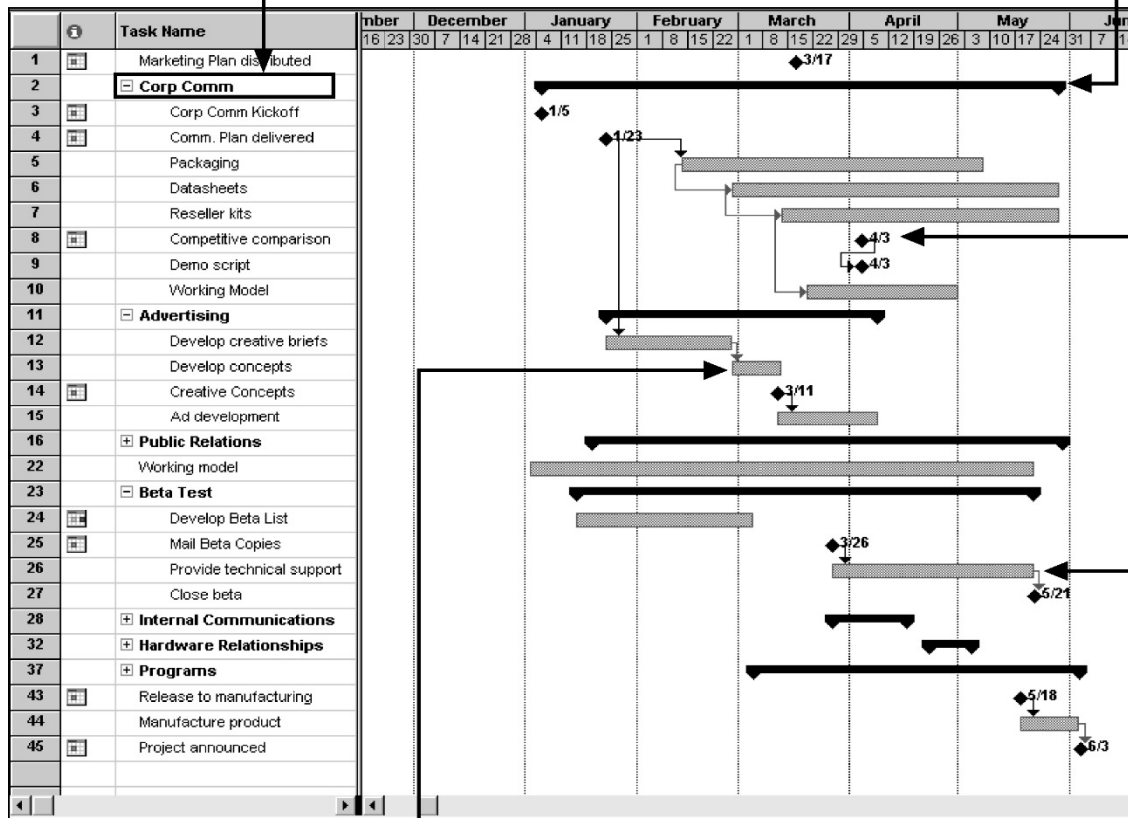
Note: Darker bars would be red in Project 2007 to represent critical tasks.



Gantt Chart for Software Launch Project

WBS hierarchy shown by indentations

Summary task



Milestone

Individual task bar

Arrows show dependencies



Adding Milestones to Gantt Charts

- Many people like to **focus on meeting milestones**, especially for large projects
- **Milestones emphasize** important **events** or **accomplishments on projects**
- Normally create milestone by **entering tasks with a zero duration**, or you can mark any task as a milestone



SMART Criteria

- **Milestones** should be:
 - Specific
 - Measurable
 - Assignable
 - Realistic
 - Time-framed

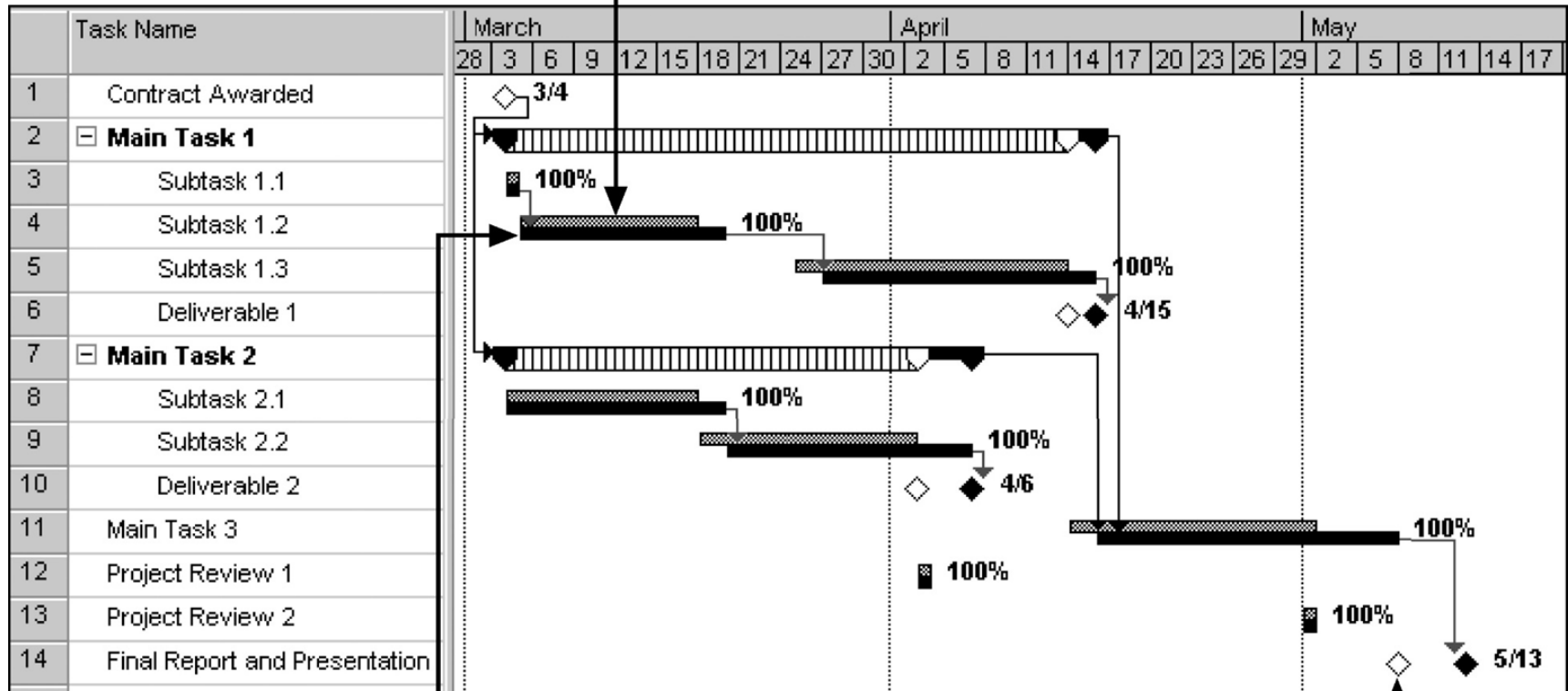


Best Practice

- **Schedule risk** is inherent in the development of complex systems. Luc Richard, the founder of projectmangler.com, suggests that **project managers can reduce schedule risk through project milestones**, a best practice that involves identifying and tracking significant points or achievements in the project.
- The **five key points** of using project milestones include the following:
 1. **Define milestones early** in the project and include them in the Gantt chart to provide a visual guide.
 2. **Keep milestones small** and frequent.
 3. The set of milestones **must be all-encompassing**.
 4. Each **milestone must be binary**, meaning it is either complete or incomplete.
 5. Carefully **monitor the critical path**.

Sample Tracking Gantt Chart

Planned dates



Actual dates

Slipped milestone



Critical Path Method (CPM)

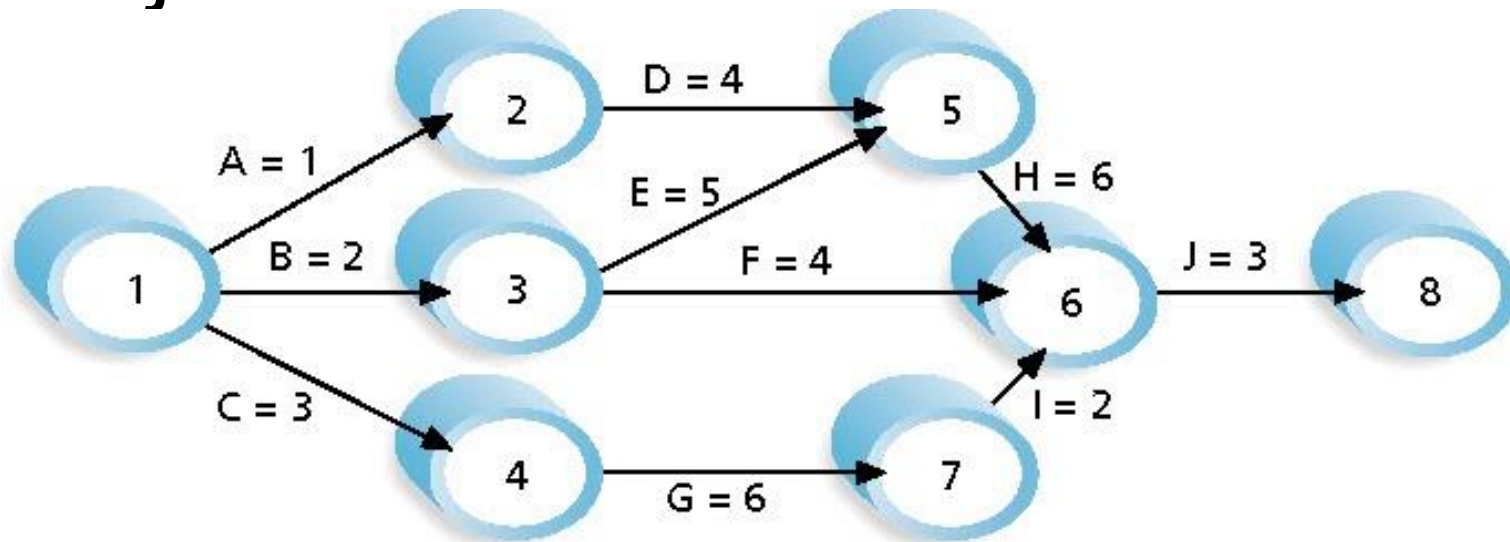
- **CPM** is a network diagramming technique used to predict total project duration
- A **critical path for a project** is the series of activities that determines the earliest time by which the project can be completed
- The **critical path** is the longest path through the network diagram and has the least amount of slack or float
- **Slack or float** is the amount of time an activity may be delayed without delaying a succeeding activity or the project finish date



Calculating the Critical Path

- First develop a **good network diagram**
- **Add the duration estimates** for all activities on each path through the network diagram
- The **longest path is the critical path**
- If one or more of the activities on the critical path takes longer than planned, the whole project schedule will slip unless the project manager takes corrective action

Determining the Critical Path for Project X



Note: Assume all durations are in days.

Path 1: A-D-H-J Length = $1+4+6+3 = 14$ days

Path 2: B-E-H-J Length = $2+5+6+3 = 16$ days

Path 3: B-F-J Length = $2+4+3 = 9$ days

Path 4: C-G-I-J Length = $3+6+2+3 = 14$ days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.



More on the Critical Path

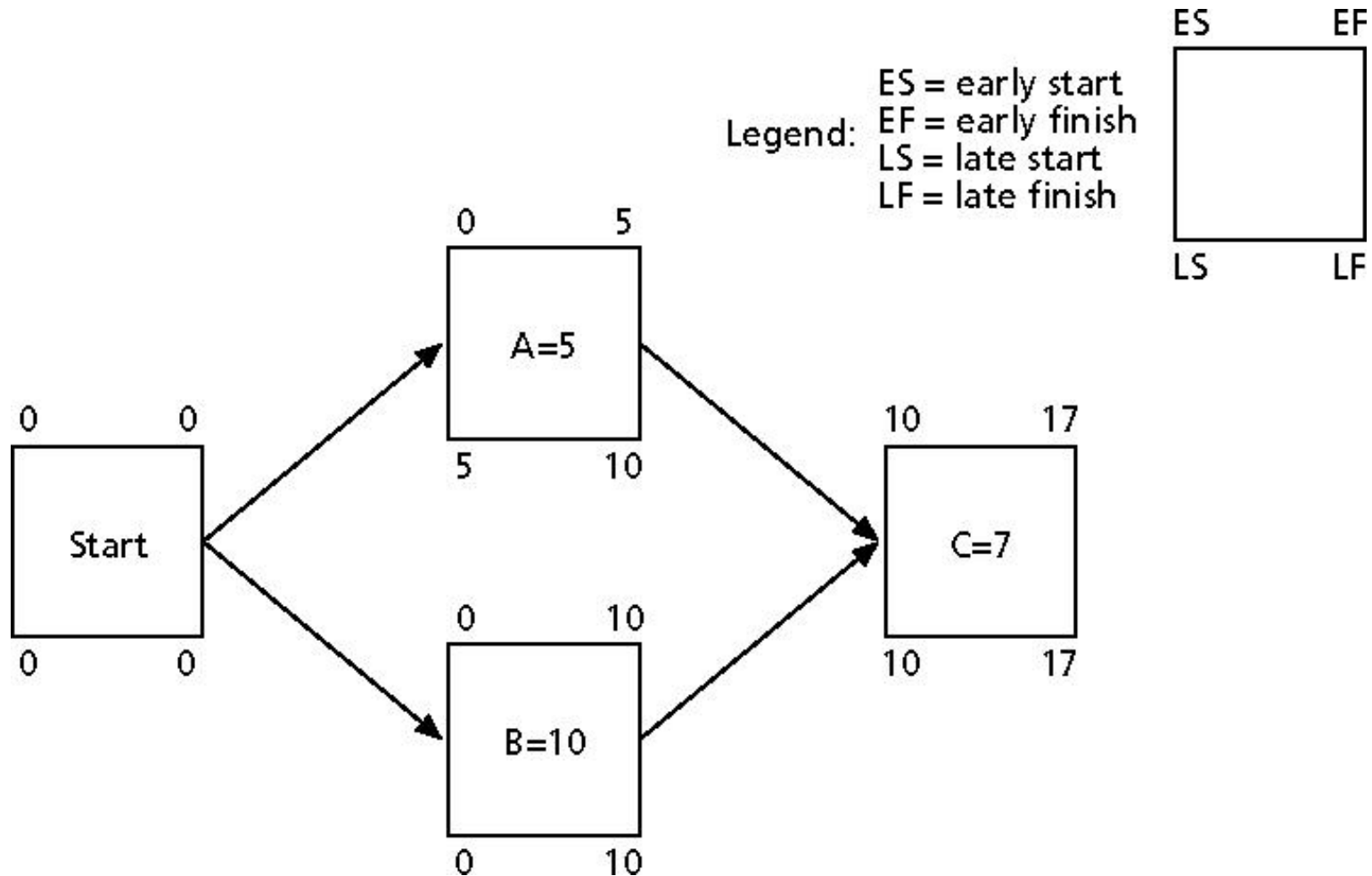
- A project team at Apple computer put a stuffed gorilla on the top of the cubicle of the person currently managing critical task
- The critical path is not the one with all the critical activities; it only accounts for time
 - Remember the example of growing grass being on the critical path for Disney's Animal Kingdom
- There can be more than one critical path if the lengths of two or more paths are the same
- The critical path can change as the project progresses



Using Critical Path Analysis to Make Schedule Trade-offs

- **Free slack** or **free float** is the amount of time an activity can be delayed without delaying the early start of any immediately following activities
- **Total slack** or **total float** is the amount of time an activity may be delayed from its early start without delaying the planned project finish date
- A **forward pass** through the network diagram determines the early start and finish dates
- A **backward pass** determines the late start and finish dates

Calculating Early and Late Start and Finish Dates

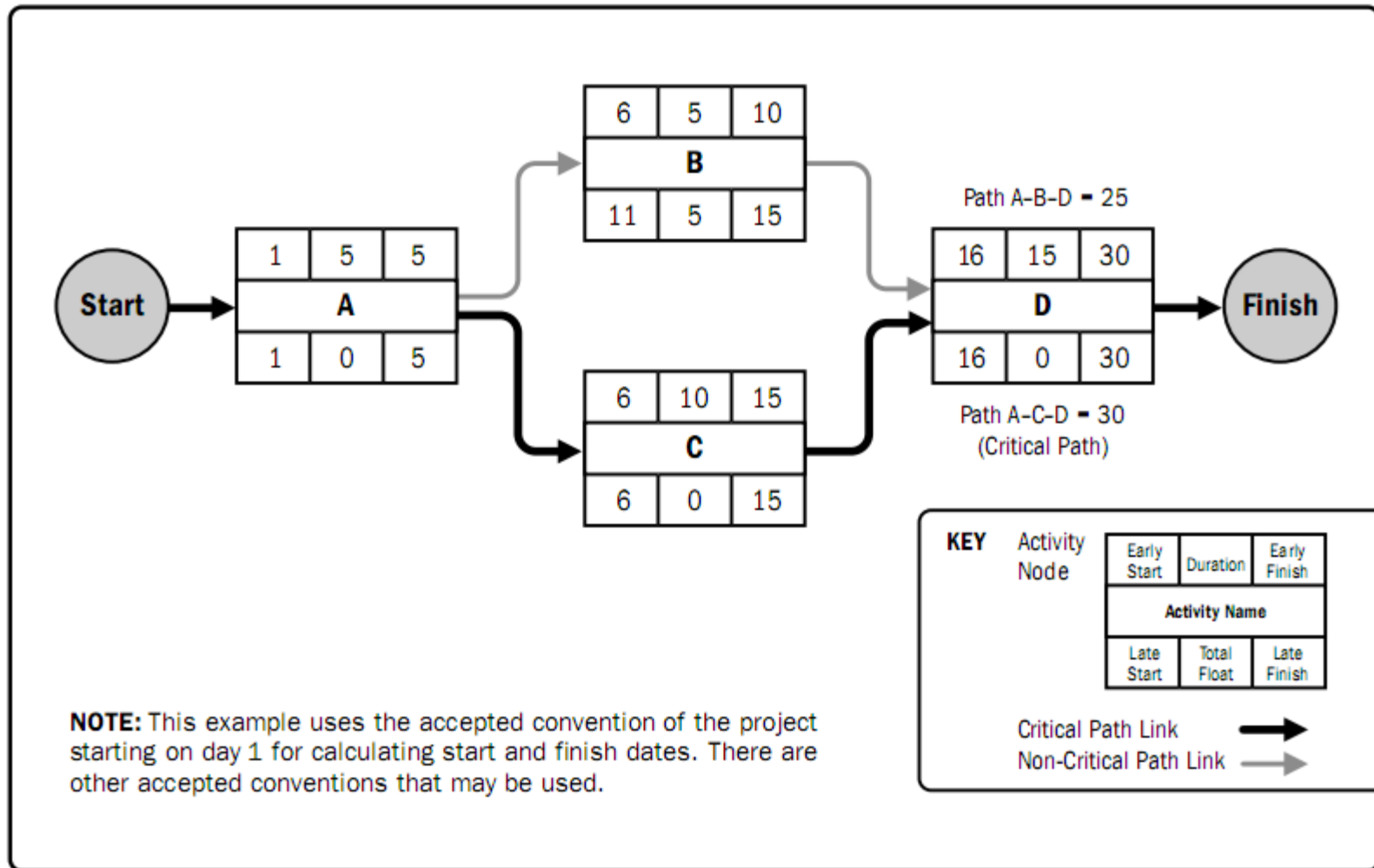




Free and Total Float or Slack for Project X

Task Name	Start	Finish	Late Start	Late Finish	Free Slack	Total Slack
A	8/3/09	8/3/09	8/5/09	8/5/09	0d	2d
B	8/3/09	8/4/09	8/3/09	8/4/09	0d	0d
C	8/3/09	8/5/09	8/5/09	8/7/09	0d	2d
D	8/4/09	8/7/09	8/6/09	8/11/09	2d	2d
E	8/5/09	8/11/09	8/5/09	8/11/09	0d	0d
F	8/5/09	8/10/09	8/14/09	8/17/09	7d	7d
G	8/6/09	8/13/09	8/10/09	8/17/09	0d	2d
H	8/12/09	8/19/09	8/12/09	8/19/09	0d	0d
I	8/14/09	8/17/09	8/18/09	8/19/09	2d	2d
J	8/20/09	8/24/09	8/20/09	8/24/09	0d	0d

Critical Path Method Example (PMBOK)





Using the Critical Path to Shorten a Project Schedule

- **Three main techniques** for shortening schedules
 1. **Shortening durations** of critical activities/tasks by adding more resources or changing their scope
 2. **Crashing activities** by obtaining the greatest amount of schedule compression for the least incremental cost
 3. **Fast tracking activities** by doing them in parallel or overlapping them



Importance of Updating Critical Path Data

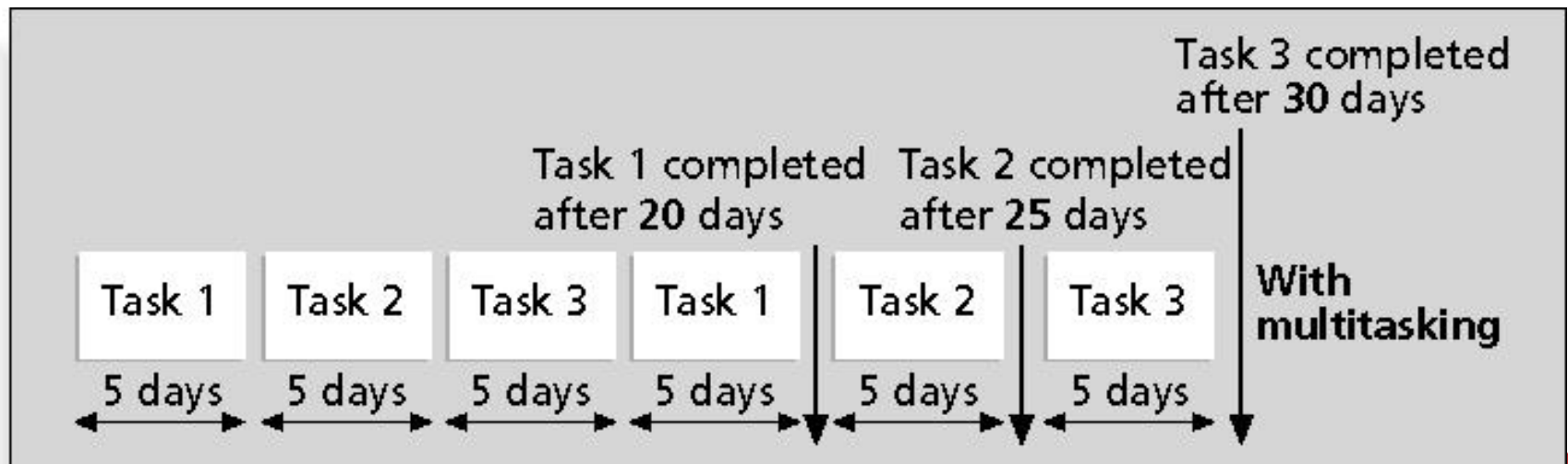
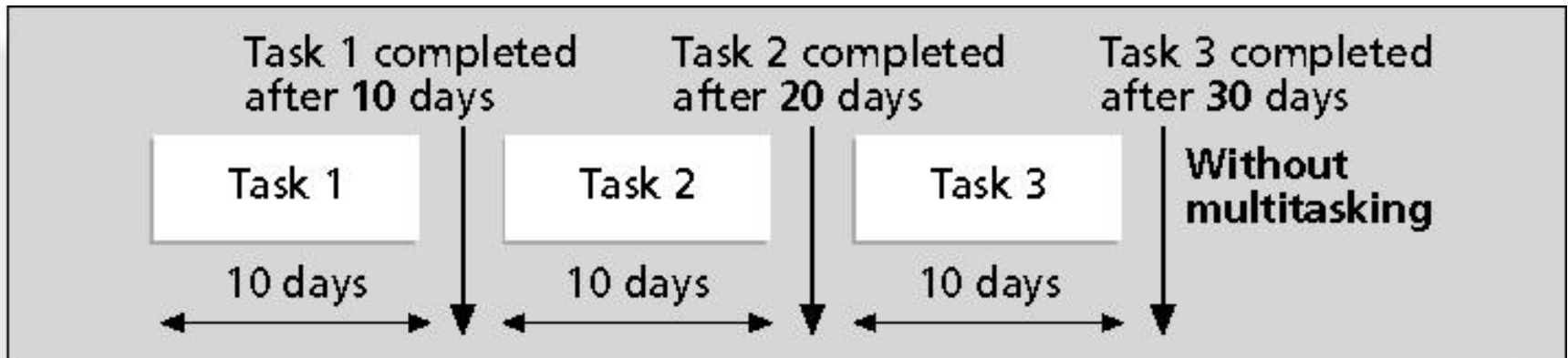
- It is **important to update project schedule information** to meet time goals for a project
- The **critical path may change** as you enter actual start and finish dates
- If you know the project **completion date will slip, negotiate with the project sponsor**



Critical Chain Scheduling

- **Critical chain scheduling**
 - A method of scheduling that considers limited resources when creating a project schedule and includes buffers to protect the project completion date
- Uses the **Theory of Constraints** (TOC)
 - A management philosophy developed by Eliyahu M. Goldratt and introduced in his book The Goal
- Attempts to minimize multitasking
 - When a resource works on more than one task at a time

Multitasking Example

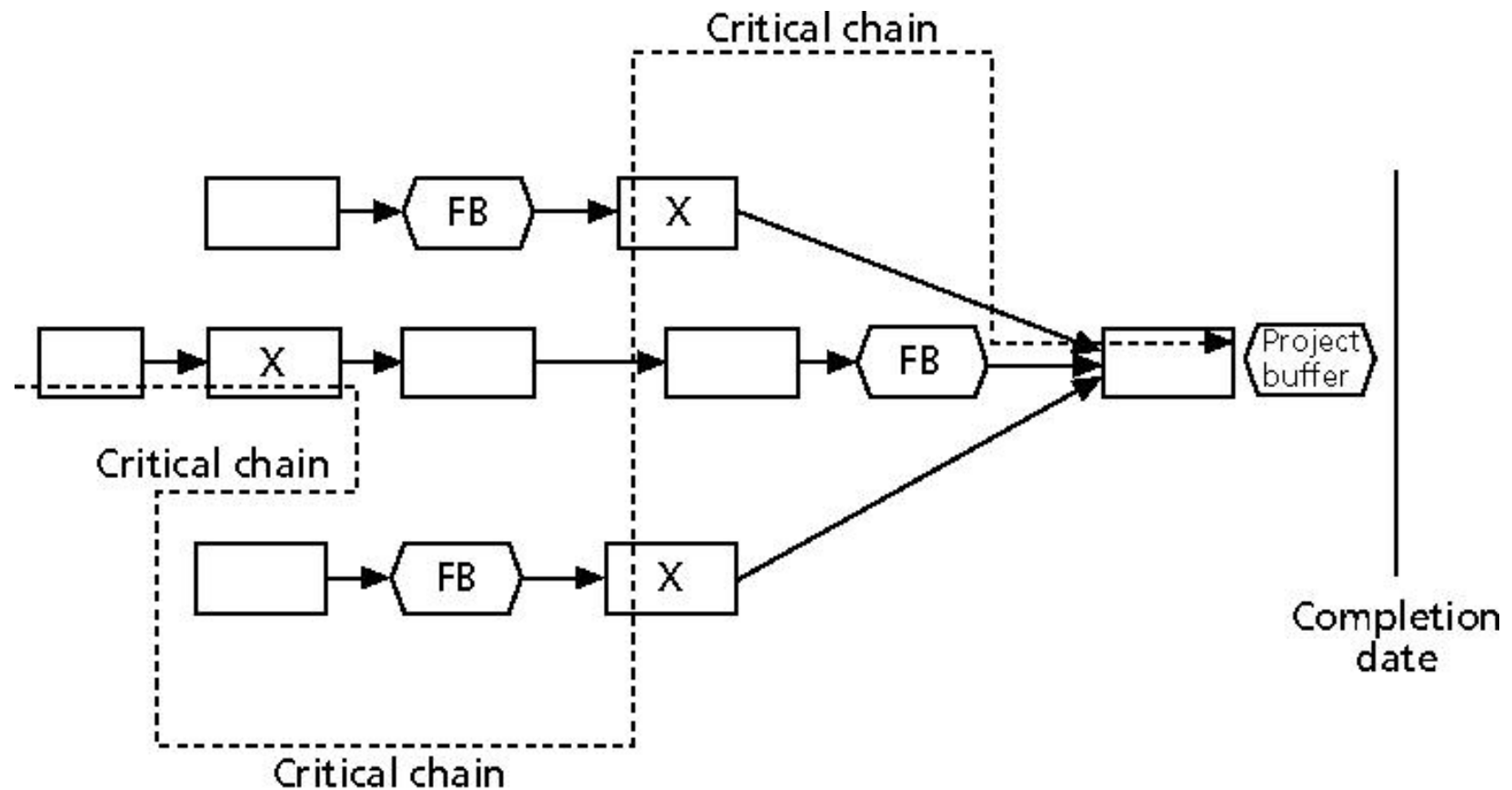




Buffers and Critical Chain

- A **buffer** is additional time to complete a task
- **Murphy's Law** states that if something can go wrong, it will
- **Parkinson's Law** states that work expands to fill the time allowed
- In traditional estimates, people often add a buffer to each task and use it if it's needed or not
- Critical chain scheduling removes buffers from individual tasks and instead creates:
 - **Project buffers** or additional time added before the project's due date
 - **Feeding buffers** or additional time added before tasks on the critical path

Example of Critical Chain Scheduling



X = Tasks done by limited resource
FB = Feeding buffer



Program Evaluation and Review Technique (PERT)

- **PERT** is a **network analysis technique used to estimate project duration** when there is a high degree of uncertainty about the individual activity duration estimates
- PERT uses probabilistic time estimates
 - Duration estimates based on using **optimistic, most likely, and pessimistic estimates** of activity durations, or a three-point estimate



PERT Formula and Example

- PERT **weighted average** =

$$\frac{\text{optimistic time} + 4 \times \text{most likely time} + \text{pessimistic time}}{6}$$

- Example:

PERT weighted average =

days $\frac{8 \text{ workdays} + 4 \times 10 \text{ workdays} + 24 \text{ workdays}}{6} = 12$

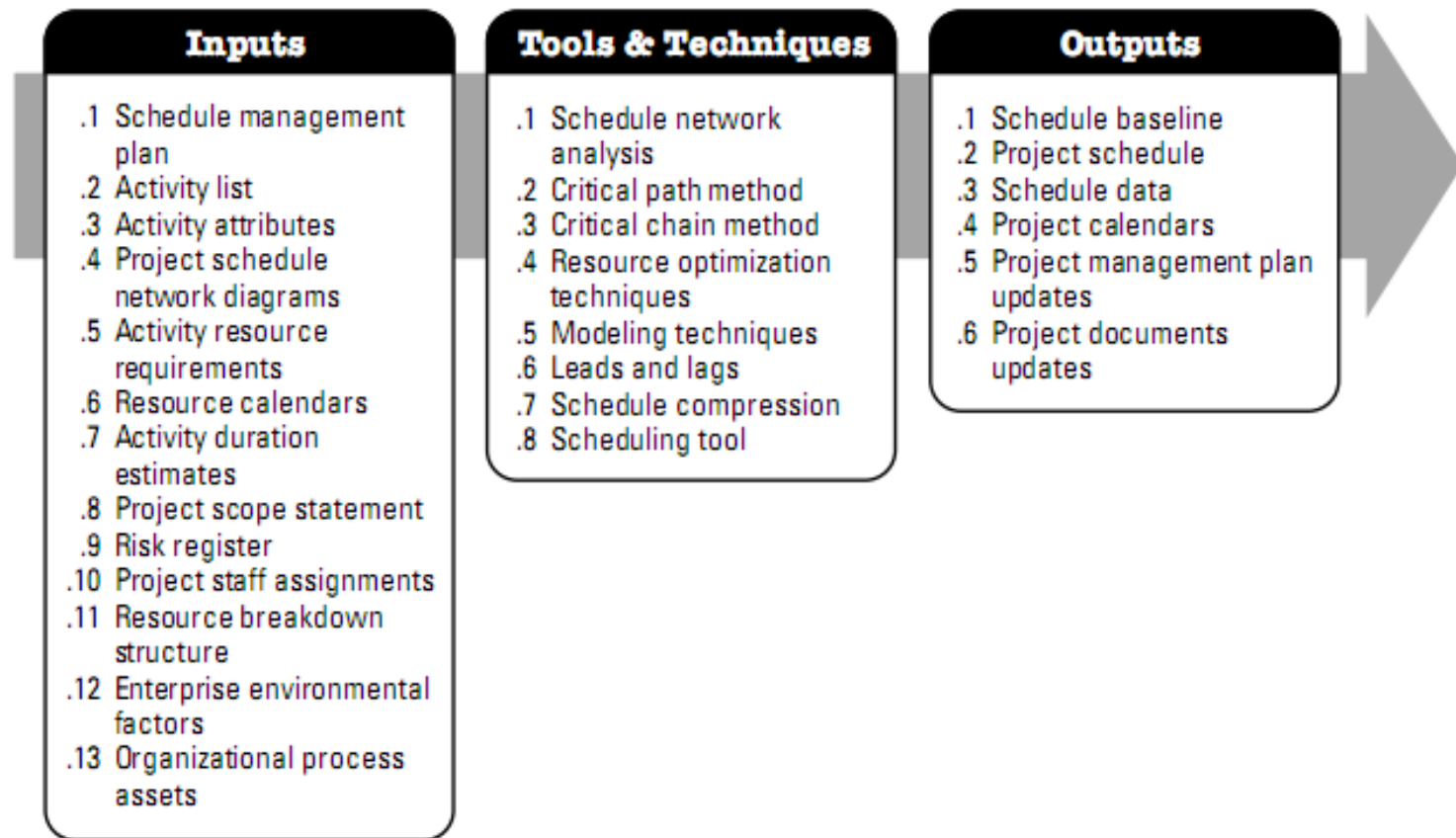
where optimistic time = 8 days

most likely time = 10 days, and

pessimistic time = 24 days

Therefore, you'd use 12 days on the network diagram instead of 10 when using PERT for the above example

Develop Schedule ITTO





Schedule Control Suggestions

- Perform reality checks on schedules
- Allow for contingencies
- Don't plan for everyone to work at 100% capacity all the time
- Hold progress meetings with stakeholders and be clear and honest in communicating schedule issues



7. Controlling the Schedule

- **Goals** are to know the **status of the schedule**, **influence factors** that cause schedule changes, **determine that** the schedule has changed, and **manage changes** when they occur
- **Tools** and techniques include:
 - **Progress** reports
 - A schedule **change control** system
 - **Project management software**, including schedule comparison charts like the tracking Gantt chart
 - **Variance analysis**, such as analyzing float or slack
 - **Performance management**, such as earned value

Control Schedule ITTO

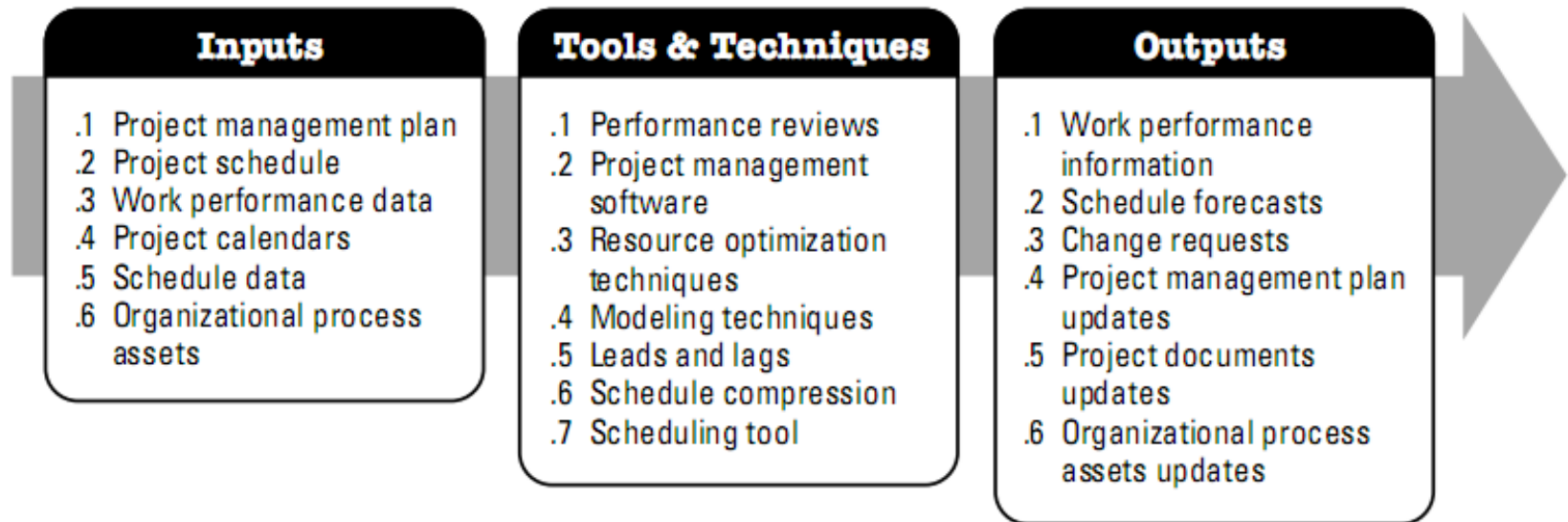


Figure 6-22. Control Schedule: Inputs, Tools & Techniques, and Outputs



Reality Checks on Scheduling

- First **review the draft schedule or estimated completion date** in the project charter
- Prepare a **more detailed schedule** with the project team
- Make **sure the schedule is realistic** and followed
- **Alert top management** well in advance **if there are schedule problems**



Working with People Issues

- **Strong leadership** helps projects succeed **more than good PERT charts**
- Project managers should use:
 - Empowerment
 - Incentives
 - Discipline
 - Negotiation



What Went Right?

- Mittal Steel Poland earned Poland's Project Excellence Award in 2007 for implementing a SAP system
- Derek Prior, research director at AMR Research, identified **three things the most successful SAP implementation** projects do to deliver business benefits:
 1. Form a **global competence center**
 2. Identify **super-users** for each location
 3. Provide **ongoing involvement of managers in business processes** so they feel they own these processes



Using Software to Assist in Time Management

- Software for facilitating communications helps people exchange schedule-related information
- Decision support models help analyze trade-offs that can be made
- Project management software can **help in various time management areas**



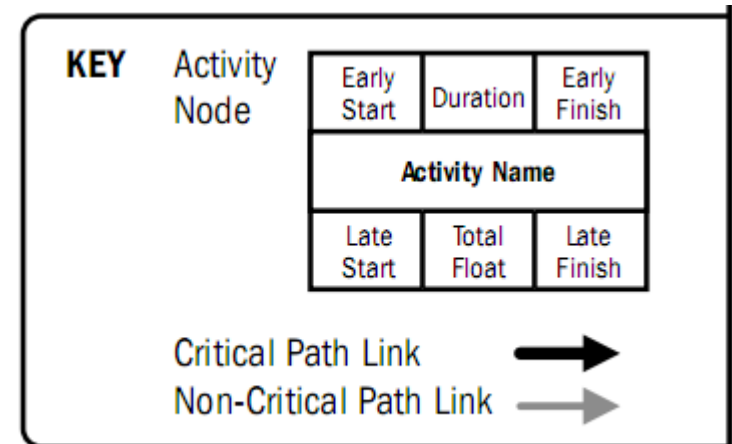
Words of Caution on Using Project Management Software

- Many **people misuse project management software** because they don't understand important concepts and have not had training
- You **must enter dependencies to have dates adjust automatically** and to determine the critical path
- You **must enter actual schedule information** to compare planned and actual progress

Exercise

- Based table on the right
 - Determine all paths and **total duration each path**
 - Determine **critical path**
 - Determine **total float** each activity
 - Calculating **Early Start** and **Early Finish** Each Activity
 - Calculating **Late start** and **Late Finish** each Activity

Name	Predecessor	Duration
Start	—	—
A	Start	6
B	A	4
C	B	8
D	A	1
E	D	1
F	E	2
G	Start	3
H	G	3
I	H	2
J	B, I	3
Finish	F, J, C	—





Summary

- **Project time management** is often **cited as the main source of conflict on projects**, and most IT projects exceed time estimates
- Main processes include:
 1. Plan Schedule Management
 2. Define activities
 3. Sequence activities
 4. Estimate activity resources
 5. Estimate activity durations
 6. Develop schedule
 7. Control schedule



References

1. Kathy Schwalbe, **Managing Information Technology Projects 7th Edition**, *Course Technology, Cengage Learning*, 2014
2. A Guide to the Project Management Body of Knowledge: **PMBOK Guide 6th Edition**, *Project Management Institute*, 2017